

Testing Documentation

This document provides evidence of successful build, test execution, and game runtime scenarios.

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Build Successful

```

MoveTest > testUppercaseConversion() PASSED
MoveTest > testMoveToString() PASSED
MoveTest > testInvalidColumn() PASSED
MoveTest > testValidMoveParsing() PASSED
MoveTest > testMoveHashCode() PASSED
MoveTest > testMoveEquality() PASSED
MoveTest > testAllColumns() PASSED
MoveTest > testNullInput() PASSED
MoveTest > testInvalidRow() PASSED

[Incubating] Problems report is available at: file:///C:/Projects/Personal/PriyaKrish_Entsia_Dots_And_Boxes/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 10.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/9.2.0/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 10s
9 actionable tasks: 9 executed
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/9.2.0/userguide/configuration_cache_enabling.html
PS C:\Projects\Personal\PriyaKrish_Entsia_Dots_And_Boxes>

```

Run Successful

```
For more on this, please refer to https://docs.gradle.org/9.2.0/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 10s
9 actionable tasks: 9 executed
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/9.2.0/userguide/configuration_cache_enabling.html
PS C:\Projects\Personal\PriyaKrish_Entsia_Dots_And_Boxes> java -cp build/classes/java/main com.entsia.dotsandboxes.DotsAndBoxes
  A B C D E F G
0 * * * *
1
2 * * * *
3
4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): |
```

Run.bat

```
Building the game...

[Incubating] Problems report is available at: file:///C:/Projects/Personal/PriyaKrish_Entsia_Dots_And_Boxes/build/reports/problems/problems-report.html
Deprecated Gradle features were used in this build, making it incompatible with Gradle 10.
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For more on this, please refer to https://docs.gradle.org/9.2.0/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 3s
8 actionable tasks: 8 up-to-date
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/9.2.0/userguide/configuration_cache_enabling.html

Starting Dots and Boxes...
  A B C D E F G
0 * * * *
1
2 * * * *
3
4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): |
```

Code Coverage

Entsia_Dots_And_Boxes [Sessions](#)

Entsia_Dots_And_Boxes

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed Cxty	Missed Lines	Missed Methods	Missed Classes
com.entsia.dotsandboxes		95%		86%	23 121	10 168	0 38	0 5
Total	39 of 863	95%	23 of 166	86%	23 121	10 168	0 38	0 5

Created with JaCoCo 0.8.13 202504020838

Testing Scenarios

1. Normal Game Flow (Happy Path)

```

4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): A1
  A B C D E F G
0 * * * *
1 |
2 * * * *
3
4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 2, input a move <column><row> (or 'Q' to quit): B2
  A B C D E F G
0 * * * *
1 |
2 * - * * *
3
4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): C3
  A B C D E F G
0 * * * *
1 |
2 * - * * *
3   |
4 * * * *
5
6 * * * *

SCORE
Player 1: 0
Player 2: 0

Player 2, input a move <column><row> (or 'Q' to quit): Q
Thanks for playing! Goodbye!
PS C:\Projects\Personal\PriyaKrish_Entsia_Dots_And_Boxes> |

```

2. Box Completion & Scoring

```

0 *  *  *  *
1 |
2 *  *  *  *
3
4 *  *  *  *
5
6 *  *  *  *

SCORE
Player 1: 0
Player 2: 0

Player 2, input a move <column><row> (or 'Q' to quit): B0
  A B C D E F G
0 * - *  *  *
1 |
2 *  *  *  *
3
4 *  *  *  *
5
6 *  *  *  *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): C1
  A B C D E F G
0 * - *  *  *
1 |  |
2 *  *  *  *
3
4 *  *  *  *
5
6 *  *  *  *

SCORE
Player 1: 0
Player 2: 0

Player 2, input a move <column><row> (or 'Q' to quit): B2
  A B C D E F G
0 * - *  *  *
1 | 2 |
2 * - *  *  *
3
4 *  *  *  *
5
6 *  *  *  *

SCORE
Player 1: 0
Player 2: 1

Player 2, input a move <column><row> (or 'Q' to quit): |

```

3. Invalid Move Rejection

```
BUILD SUCCESSFUL in 2s
8 actionable tasks: 8 up-to-date
Consider enabling configuration cache to speed up this build:
ml

Starting Dots and Boxes...

  A B C D E F G
0 *  *  *  *  *
1
2 *  *  *  *
3
4 *  *  *  *
5
6 *  *  *  *

SCORE
Player 1: 0
Player 2: 0

Player 1, input a move <column><row> (or 'Q' to quit): Z2
Invalid move. Please try again.
Player 1, input a move <column><row> (or 'Q' to quit): |
```

4. Position already Occupied

```

Player 1, input a move <column><row> (or 'Q' to quit): C1
  A B C D E F G
0 * - *   *   *
1 |   |
2 *   *   *   *
3
4 *   *   *   *
5
6 *   *   *   *

SCORE
Player 1: 0
Player 2: 0

Player 2, input a move <column><row> (or 'Q' to quit): B2
  A B C D E F G
0 * - *   *   *
1 | 2 |
2 * - *   *   *
3
4 *   *   *   *
5
6 *   *   *   *

SCORE
Player 1: 0
Player 2: 1

Player 2, input a move <column><row> (or 'Q' to quit): B2
This position is already occupied. Please try again.
Player 2, input a move <column><row> (or 'Q' to quit): |

```

5. Winner Declaration

```

Player 2, input a move <column><row> (or 'Q' to quit): G3
  A B C D E F G
0 * - * - * - *
1 | 2 | 1 | 2 |
2 * - * - * - *
3 | 2 | 1 | 2 |
4 * - * - * - *
5 | 2 | 1 |
6 * - * - * - *

SCORE
Player 1: 3
Player 2: 5

Player 2, input a move <column><row> (or 'Q' to quit): G5
  A B C D E F G
0 * - * - * - *
1 | 2 | 1 | 2 |
2 * - * - * - *
3 | 2 | 1 | 2 |
4 * - * - * - *
5 | 2 | 1 | 2 |
6 * - * - * - *

SCORE
Player 1: 3
Player 2: 6

Game over. Player 2 is the winner!
Thanks for playing! Goodbye!
Press any key to continue . . . |

```


6. Quit Game

```
Starting Dots and Boxes...
```

```
  A B C D E F G
0 *   *   *   *
1
2 *   *   *   *
3
4 *   *   *   *
5
6 *   *   *   *
```

```
SCORE
```

```
Player 1: 0
```

```
Player 2: 0
```

```
Player 1, input a move <column><row> (or 'Q' to quit): Q
```

```
Thanks for playing! Goodbye!
```

```
Press any key to continue . . . |
```